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Smart Shopping Cart

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TARGETS OF THE MINI PROJECT WORK

ACTIVITIES		ACTIVITIES	
Domain & problem identified	Yes	Development of product	No
Literature Review	Yes	Testing	No
Objectives formulated	Yes	Obtained Result	No
Methodology / Design	Yes	Documentation	No
Created work plan and task allocation	Yes	Report submission	No

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INTRODUCTION

- Smart Shopping Cart aims to enhance the shopping experience by reducing time spent at billing counters.
- Traditional billing relies on barcode scanning, requiring manual effort and increasing checkout time.
- The proposed system uses RFID tags instead of barcodes to streamline the shopping process.



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INTRODUCTION

- Uses an RFID reader with an LCD display mounted on the shopping cart to display the price of each scanned item.
- Improves overall customer satisfaction and operational efficiency in shopping malls.



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OBJECTIVES

- Develop an RFID-based smart shopping cart to automate item detection and displaying item details and total cost in real time.

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EXISTING SYSTEM

- The existing system relies on barcode scanning, requiring each item to be scanned manually at the checkout counter.
- Manual scanning in traditional billing systems increases the chances of human errors, such as scanning the wrong item, multiple scans, or missed items, resulting in inaccurate billing.



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EXISTING SYSTEM

- The dependency on barcode scanners means that damaged or missing barcodes can cause delays and require manual price entry.
- Customers need to remove items from their carts for scanning, which adds inconvenience and increases the overall shopping time.

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LITERATURE REVIEW

SL NO	Name of the Paper	Outcome	Author and Publication Details
1	Data transmission using RFID system on smart shopping carts for checkout process efficiency in supermarket at indonesia	- The project resulted in the successful implementation of a Smart Shopping System, which reduced queue times by allowing customers to self-scan items, thus making the checkout process more efficient and convenient. - Customers need to familiarize themselves with the self-scanning feature and smartphone app, including scanning items, checking prices, and managing shopping lists. Some may face a learning curve while adapting to this technology.	Martinusa , Metta Saridewi Wahaba , Yudia, Hanry Ham, published in the journal Procedia Computer Science, volume 179 on January 1, 2021

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SL NO	Name of the Paper	Outcome	Author and Publication Details
2	Smart shopping cart using RFID technology	- In this system, the customer begins by logging in via a QR code on the shopping cart, enabling a seamless shopping experience. The cart automatically detects items using RFID technology, updates the bill in real-time, and manages stock efficiently. - A challenge in this system is ensuring user-friendly QR code login process, as some customers may face difficulties with scanning, app compatibility, or network issues, potentially delaying their shopping experience.	Susanna M Santhosh, Cemone P Babu, Jills Joseph , Preethu Benny, published in the International Journal of Engineering Applied Sciences and Technology (IJEAST), Volume 5, on June 2020.

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SL NO	Name of the Paper	Outcome	Author and Publication Details
3	IOT based Smart Shopping Cart using RFID	• This system utilizes RFID technology, an Arduino microcontroller, and a Bluetooth module to track items in real time and automate billing. It reduces checkout times, provides product details, and updates inventory by recording purchases and generating bills automatically. • The main challenges faced in this project include high initial setup costs, customer adaptation to new technology, ensuring data privacy and security, and the need for regular maintenance and updates.	Seshagopal , Harish D , Ashwinth Fatin , Dr. Antonidoss , published in the International Journal of Engineering Research & Technology (IJERT), Volume 10, Issue 04, in April 2021.

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SL NO	Name of the Paper	Outcome	Author and Publication Details
4	RFID Based Smart Shopping Cart for Smart Shopping Enhanced with IoT	- This RFID-based system automates billing by scanning tagged products and displaying details on an LCD screen. It reduces wait times and uses GSM modules for real-time updates and notifications. - A challenge in this system is maintaining stable GSM connectivity, as network issues or poor signal strength may disrupt real-time, affecting the efficiency of the checkout process	K Sreenivasa Rao, Ayesha Shaik, N Dineshsairamreddy, H Anusha, A Guru Rajesh, published in the Journal of Emerging Technologies and Innovative Research (JETIR), Volume 10, Issue 3, in March 2023.

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SL NO	Name of the Paper	Outcome	Author and Publication Details
5	Research on Smart Shopping Cart	• This project develops an IoT-based Smart Shopping Cart using RFID technology to automate billing and enhance the shopping experience. It features automatic item detection, reduces checkout wait times. The system integrates an microcontroller, GSM module, LCD, and keypad to display costs. • challenge for this system is ensuring seamless communication between the GSM module and the server, as network disruptions or delays could affect real-time transaction updates and payment notifications, leading to a less efficient shopping experience.	Prof. Roopa C , Nivas Chandra Reddy, published in the Journal of Emerging Technologies and Innovative Research (JETIR), Volume 11, Issue 3, in March 2024.

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PROPOSED SYSTEM

- RFID Scanning: Scan items as you add them to the cart for instant recording.
- Real-Time Display: LCD screen displays item details and updated totals.
- Automated Billing: Cart maintains a live total for fast and easy checkout.
- No Manual Scanning: Eliminates manual barcode scanning, reducing cashier workload and wait times.



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ADVANTAGES

- Reduced Checkout Stress
- Efficient Multi-Item Scanning
- Increased Customer Satisfaction

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METHODOLOGY

Module 1: RFID-Based Product Identification Module

Objective: Automatically detect and identify products added to the cart

Components: RFID Tags → RFID Reader → Microcontroller

Module 2. Billing & Pricing Module

Objective: Maintain an updated total bill as products are added/removed.

Components: Microcontroller → Display Unit

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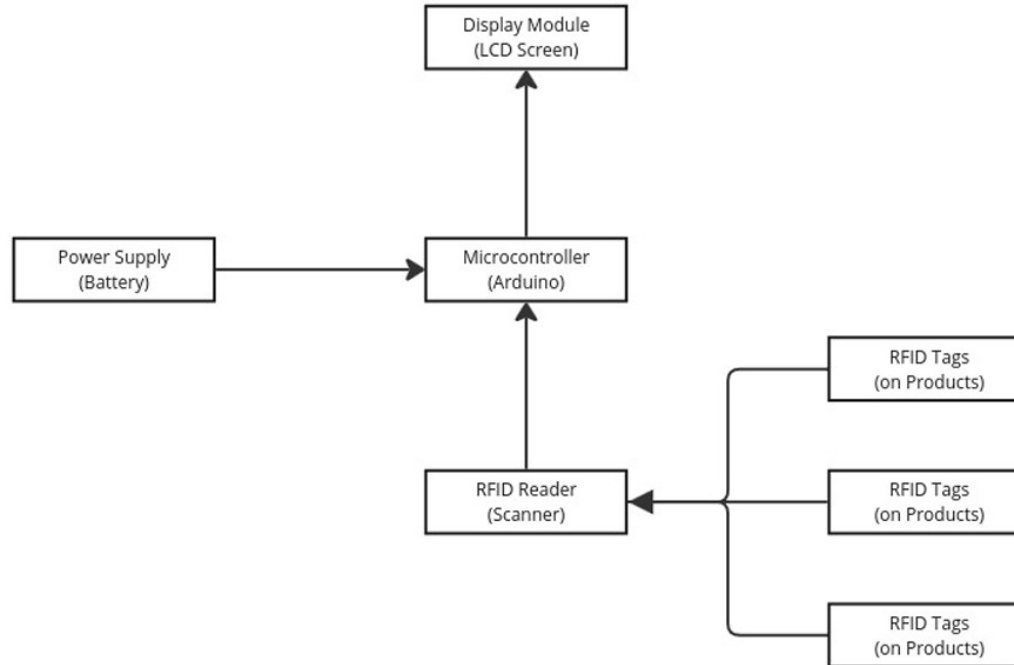
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BLOCK DIAGRAM



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RESOURCES NEEDED

SOFTWARE REQUIREMENTS:

- Arduino IDE

HARDWARE REQUIREMENTS:

- RFID Reader (Mounted on the cart)
- RFID Tags (Attached to products)
- Microcontroller (Arduino)
- Display Unit (LCD)
- Power Supply (Rechargeable battery)

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WORK PLAN

ABIN SANTHOSH	Development & Implementation
ADHITH SUNIL	Project Monitoring & Quality Check
ADHWAITH T T	Testing & Risk Management
ANITTA RAPHI E	Documentation & Resource Handling



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FEASIBILITY ANALYSIS

- Technical Feasibility
- Economic Feasibility
- Social Feasibility
- Legal & Ethical Feasibility

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CONCLUSION

- Before implementation, we structured our work by setting a work plan.
- We defined the problem statement and finalized the methodology.
- Additionally, we procured the necessary components to carry out the project effectively.
- We purchased the necessary components.

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- [2] Susanna M Santhosh, Cemone P Babu, Jills Joseph , Preethu Benny, published in the International Journal of Engineering Applied Sciences and Technology (IJEAST), Volume 5, on June 2020.
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[5] Prof. Roopa C , Nivas Chandra Reddy, published in the Journal of Emerging Technologies and Innovative Research (JETIR), Volume 11, Issue 3, in March 2024.

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